

JOYITA FARUK

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RESEARCH INTERESTS

Computational Biology, AI/Deep Learning, Multi-modal Learning, Graph Neural Networks, Data Analytics, Quantum Computing

EDUCATION

Bachelor of Science | *Computer Science* | *CGPA 3.54/4.0*
Independent University, Bangladesh

Sep. 2019 – Aug 2024
Dhaka, Bangladesh

WORK EXPERIENCE

Junior Data Scientist, R&D Department
SimpliSolve

Jun 2025 – Current
Texas, US (Remote)

- Researching the latest methodologies to support the development of R&D projects
- Understanding and developing state-of-the-art AI models
- Conducting thorough exploratory data analysis, discerning patterns and insights
- Writing efficient and robust GPU-compatible Python code
- Deploying developed codes using cloud services such as AWS
- Spearheading an AI Innovators talent hunt at Independent University Bangladesh on behalf of the company, designing the competition dataset, coordinating logistics by collaborating with marketing and academic teams.

Genomics Intern
Genereveal

Feb 2025 – Jun 2025
Montana, US (Remote)

- Worked on human DNA sequence data, exploring disease associations
- Leveraged Slurm-based HPC for large-scale genomic data processing
- Analyzed genomic data via alignment, variant calling, and tools like BWA, GATK, and SAMtools
- Developed and optimized bash scripts to automate workflows

RESEARCH

Kidney Abnormality Detection Using Segmentation-Guided Classification on CT Images

Published in: International Conference on Machine Learning and Cybernetics (ICMLC), Miyazaki, Japan.
Pages 414–419. DOI: 10.1109/ICMLC63072.2024.10935113

Kidney Abnormal Region Localization on 3D CT Images Using ROI-Guided Segmentation

Thesis project. Journal paper in progress

SKILLS

Machine Learning & AI: Keras, TensorFlow, PyTorch, AI Agent Workflows

Cloud & DevOps: AWS, Docker, Terraform, Linux Shell Scripting

Programming Languages: Python, Java, C++

Database: MySQL, Firebase, Neo4j

Web & Software Development: Django, Flutter

Document Creation: $\LaTeX$, Markdown, Microsoft Office Suite

Languages: Bengali (Native), English (IELTS Academic 8.5), French (Preparing for A1)

HONORS AND AWARDS

Mitacs Globalink Research Award

Competitive award for conducting research internship in Canada; Grant Amount: \$6000 CAD

January 2025

3-Minute Thesis Competition Winner

1st Place at the CSE Annual Research Day, Independent University, Bangladesh

March 2024

Microprocessor Project Showcase Champion

1st Place at IUB Autumn Tech Fest, recognized for technical innovation and presentation

December 2023

Web Application Project Showcase Winner

1st Place at IUB Summer Tech Fest for developing an innovative competition platform prototype

July 2023

Dean's List

Awarded for academic excellence at Independent University, Bangladesh

2023

Dean's Merit List

Recognized for being among the top-performing students in the School of Computer Science

2022

Merit Scholarship

Awarded based on university entrance exam; received full or partial tuition coverage across multiple semesters

2019–2022

WORKSHOPS

Interdisciplinary Computational Biology Workshop 2025

3-Day Workshop at Brac University

January 2025
Dhaka, Bangladesh

2nd CCDS Workshop on Deep Learning Code Management

2-Day Workshop by CCDS, Independent University, Bangladesh

December 2024
Dhaka, Bangladesh

COURSE PROJECTS

TakaTracker – Expense Tracking Mobile Application

Developed using Flutter and Firebase with features like currency conversion and categorized expense tracking

2024

Cloud HemoSphere – Cloud-Based Blood Banking Web App

Hosted on AWS using EC2 and VPC to ensure data security for blood bank management

2023